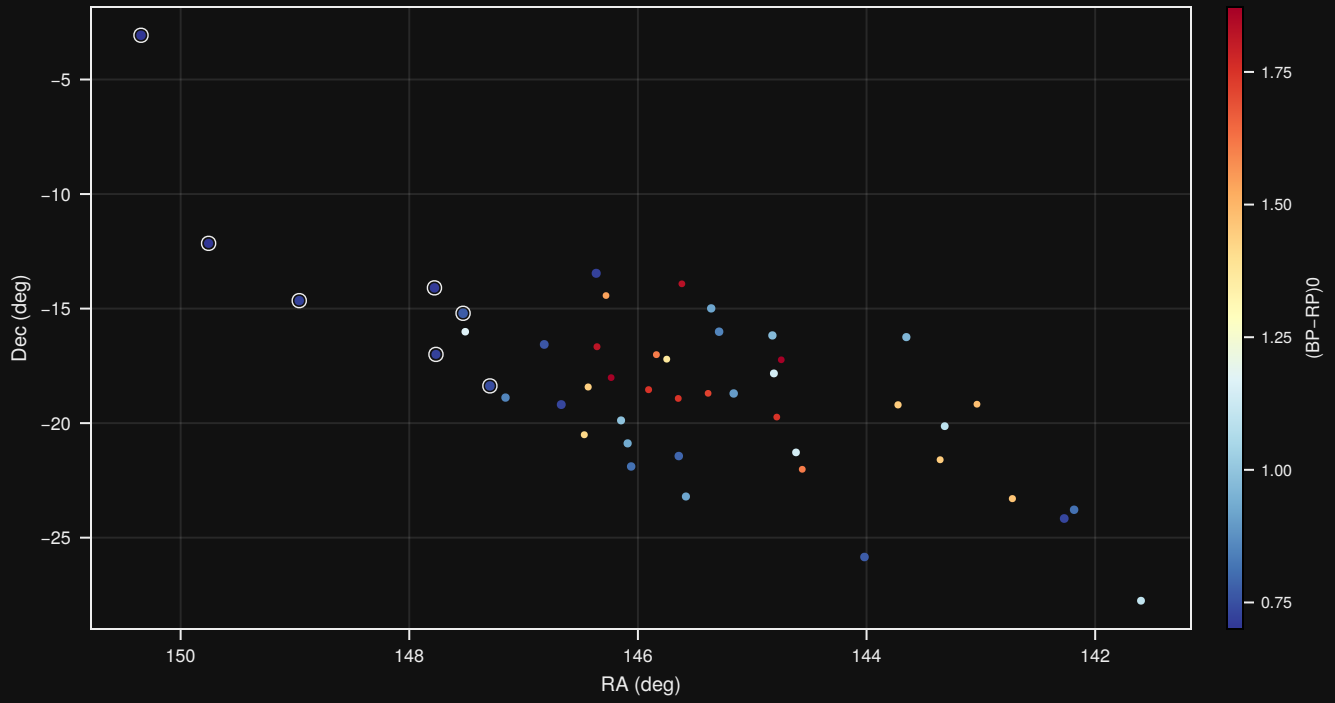
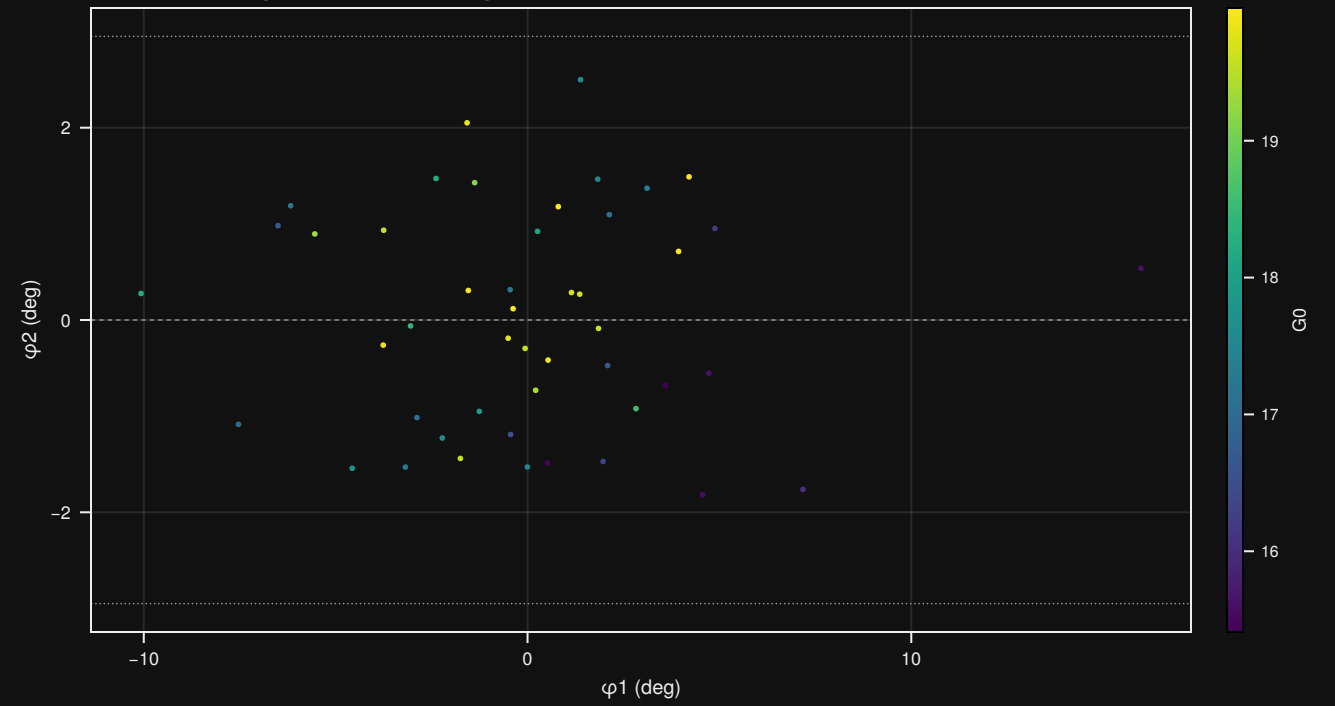
New-10

MEMBERSHIP	
Stars in catalog	47
with Gaia GSP-Phot	28 (60%)
with radial velocity	7
Stream id (sID)	31
GEOMETRY · great-circle fit	
Centroid RA, Dec	145.6, -18.5 deg
Centroid l, b	252.4, 25.5 deg
Pole RA, Dec	61.9, 18.3 deg
Length (φ1, 2–98%)	26.1 deg
Width σ(φ2), robust	1.48 deg
KINEMATICS	
pmRA* med ± σ	2.14 ± 0.56 mas/yr
pmDE med ± σ	6.81 ± 0.71 mas/yr
μφ1* med ± σ	7.25 ± 0.61 mas/yr
μφ2 med ± σ	0.16 ± 0.56 mas/yr
Parallax (cat.) med	0.426 ± 0.139 mas
Parallax (wtd, ZP-corr.)	0.439 ± 0.016 mas
Distance	d = 2.3 ± 0.1 kpc
Physical length	1.04 kpc
RV median ± σ(MAD)	111.6 ± 2.0 km/s
RV gradient dV/dφ1	-0.08 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.4–20.0 (17.9)
(BP−RP)0 median	0.99
GSP-Phot [M/H] (n=28)	-1.30 ± 0.27
GSP-Phot Teff median	5398 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (14 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	47 | 47
n_RV catalog | paper	7 | 7
<[Fe/H]> paper	-0.99
Stellar mass (paper)	100 Msun
Pole offset vs paper	1.2 deg

Sky map · catalog members (n = 47) · size ∝ brightness · ring = has RV

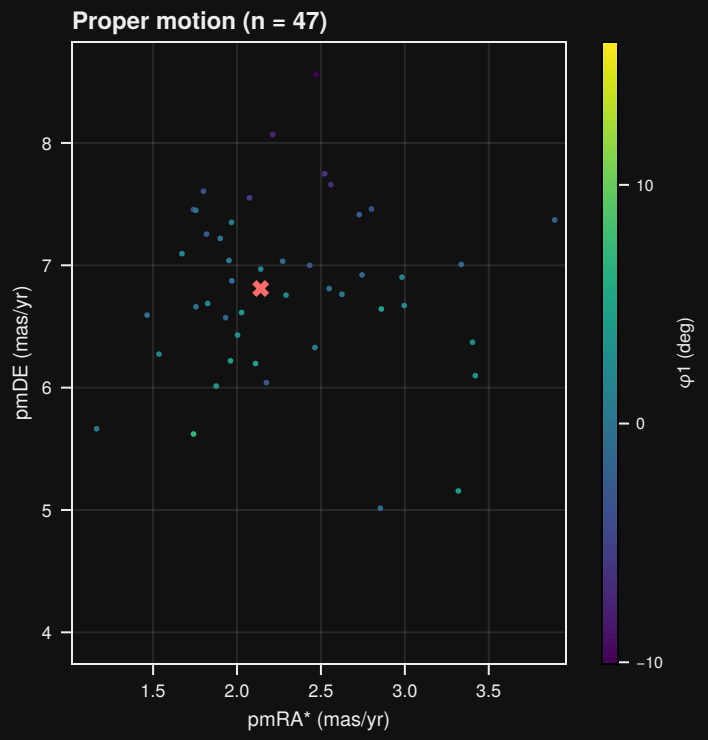


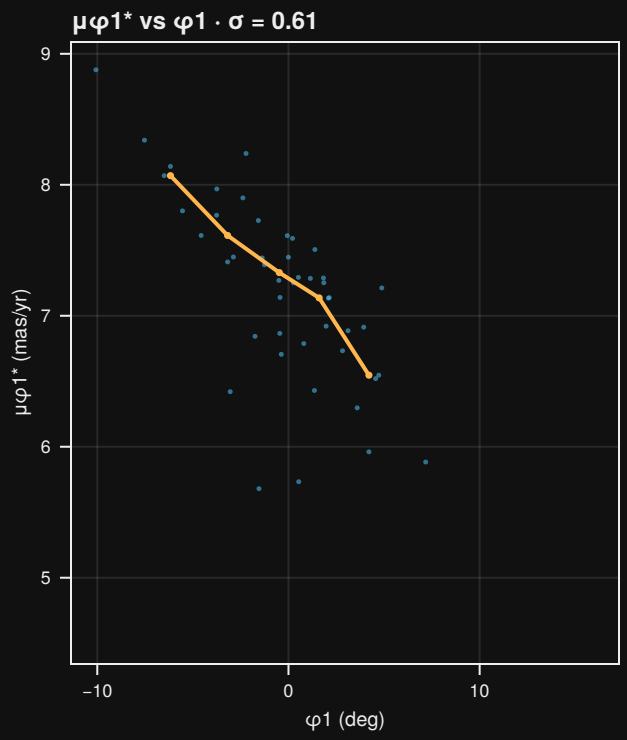
Stream frame (great-circle fit) · length 26.1° · width σ 1.48°

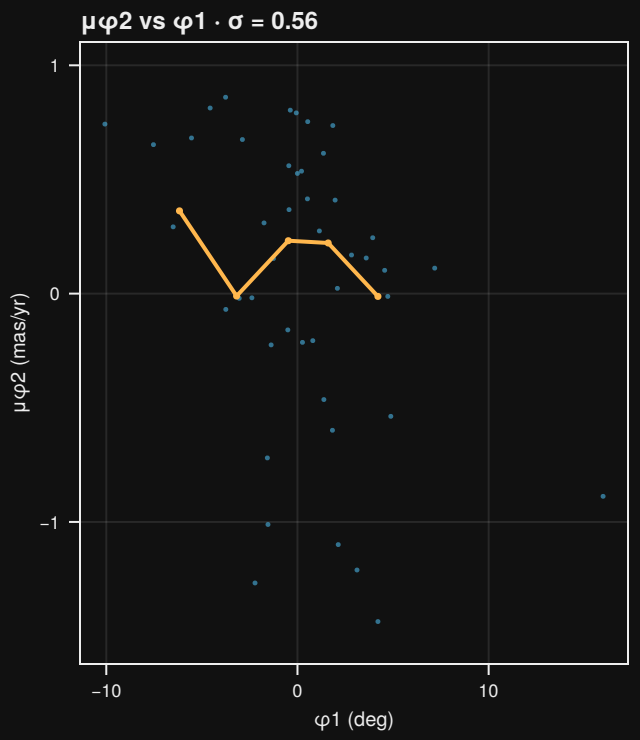


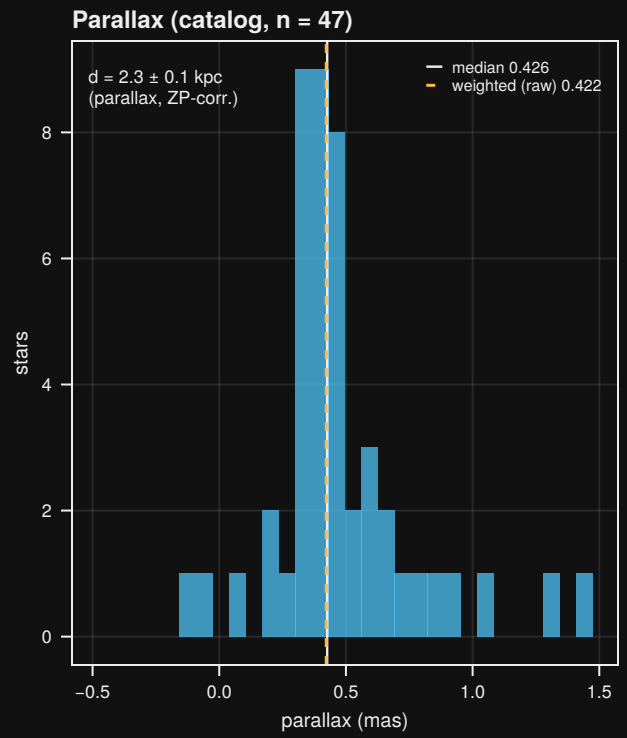
New-10

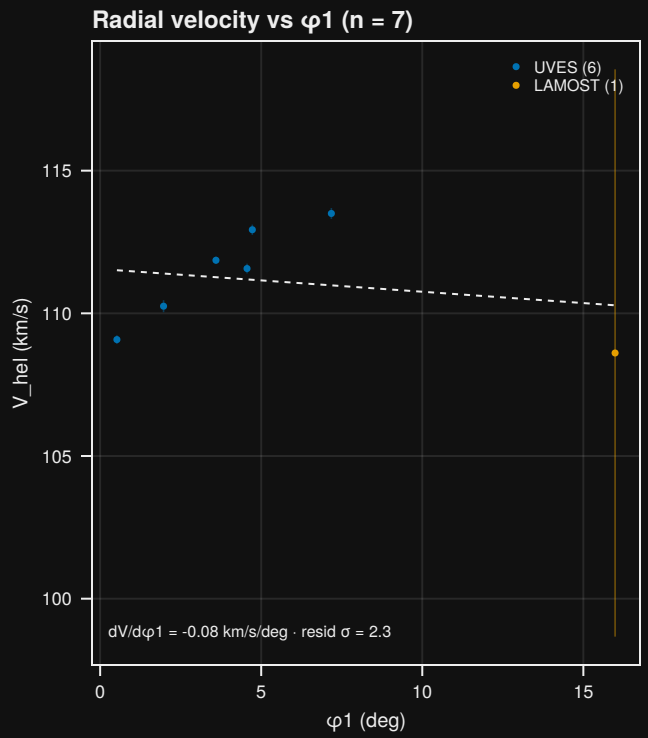
Kinematics · proper motion · parallax · radial velocity

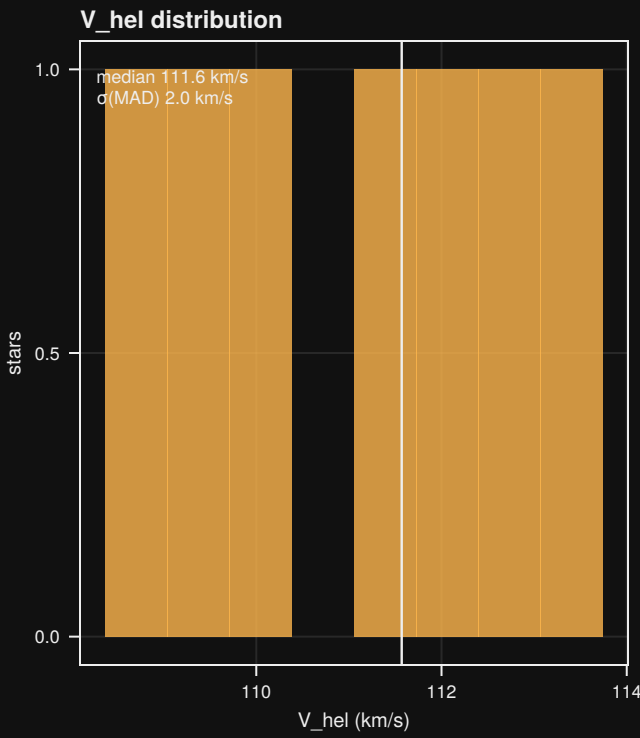






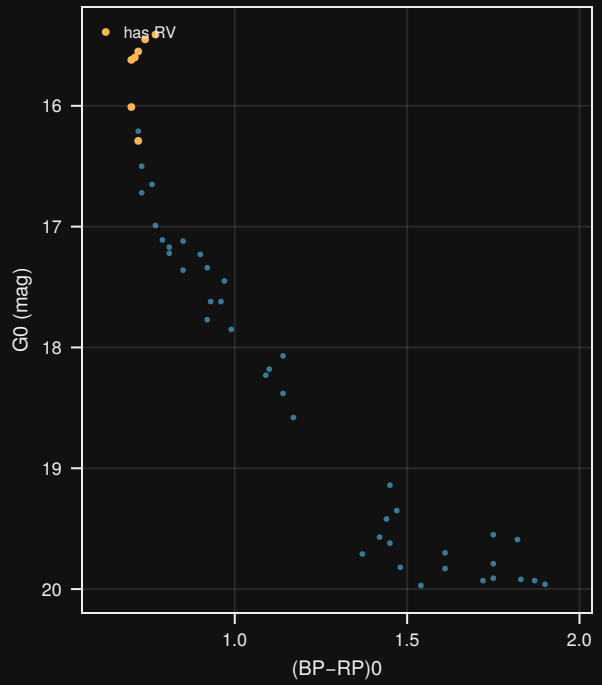




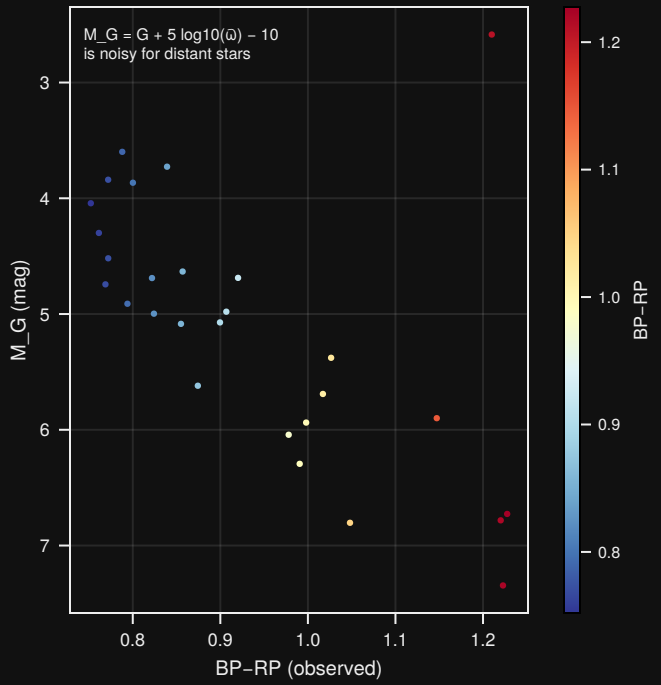


New-10

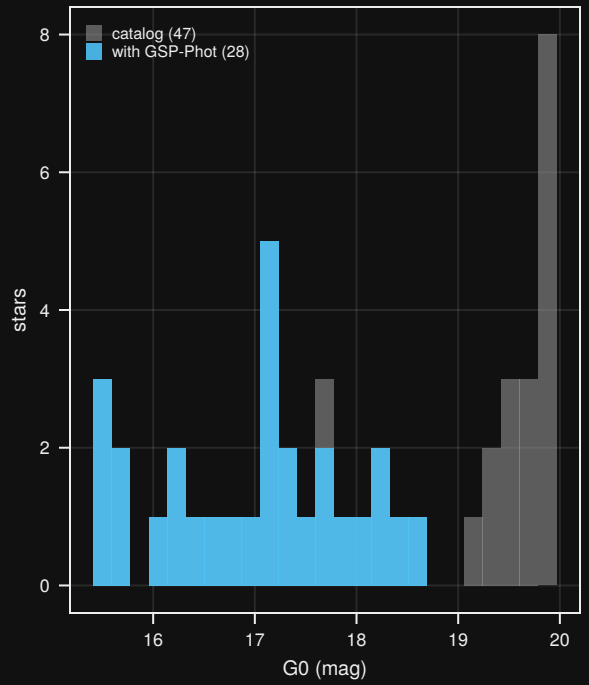
CMD · dereddened (n = 47)



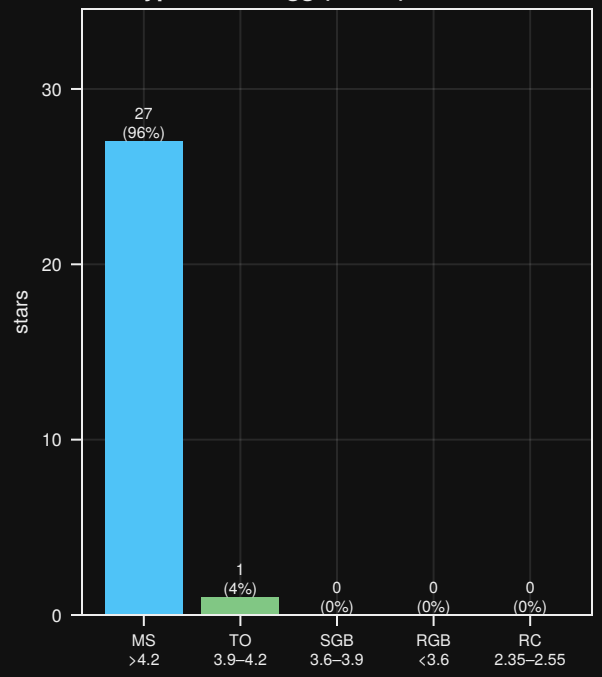
HR diagram · ϖ>0 (n = 28/28)



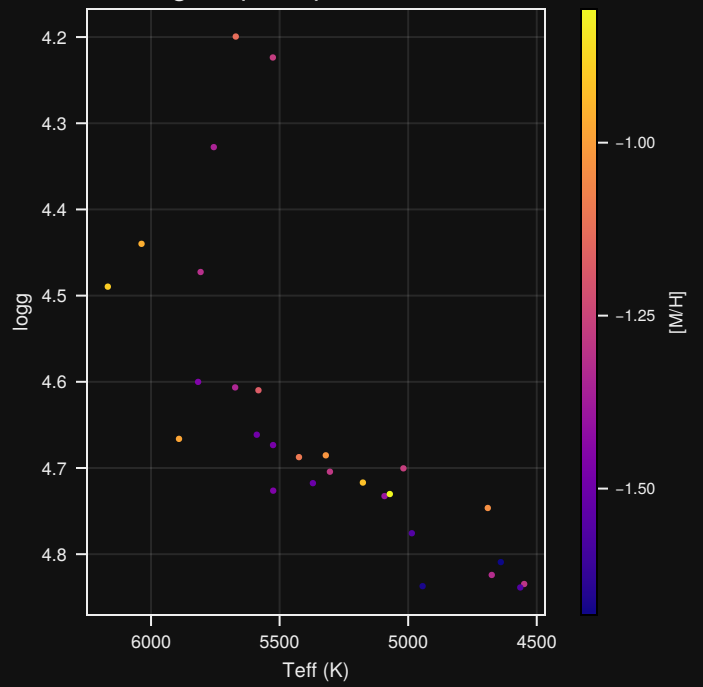
G0 & GSP-Phot coverage



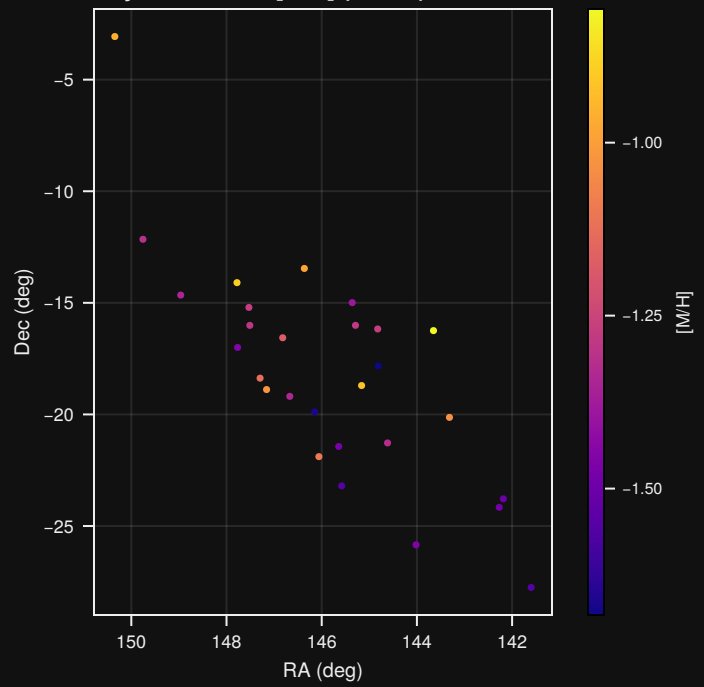
Star types from logg (n = 28)



Kiel diagram (n = 28)



Sky · GSP-Phot [M/H] (n = 28)

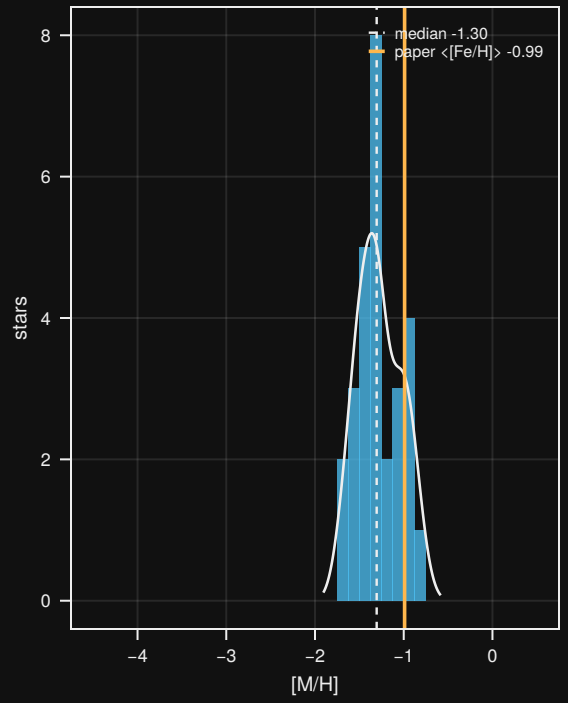


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

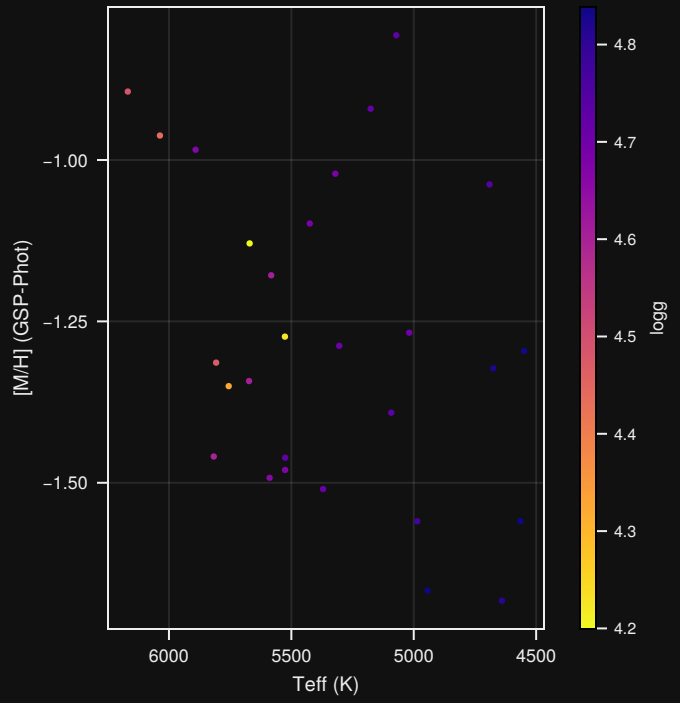
New-10

Metallicity · age · astrometric quality

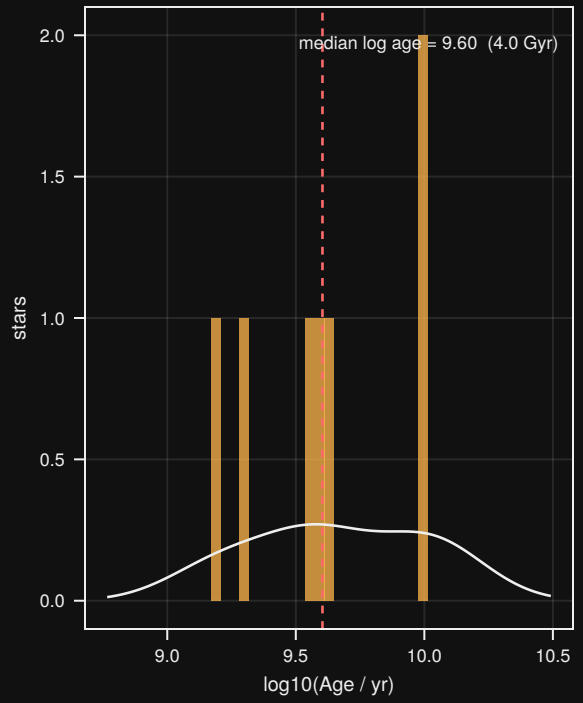
[M/H] GSP-Phot (n = 28)



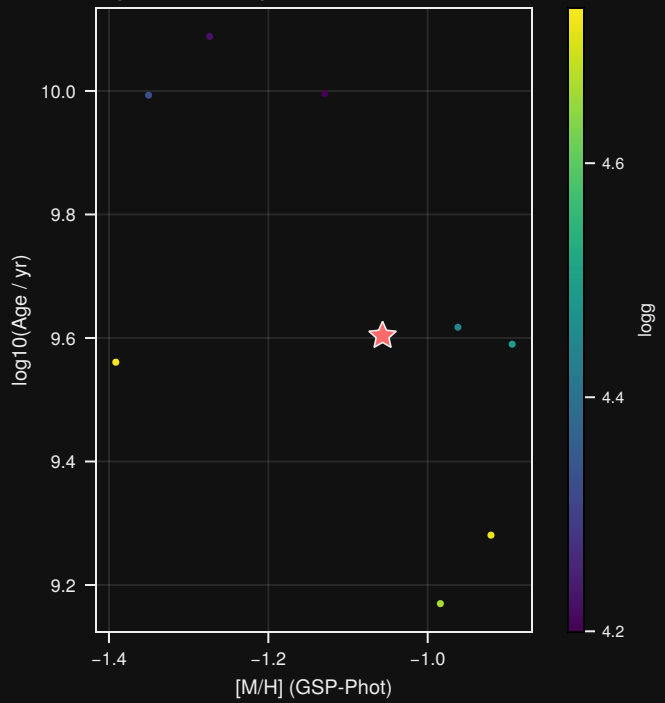
Teff vs [M/H] (n = 28)



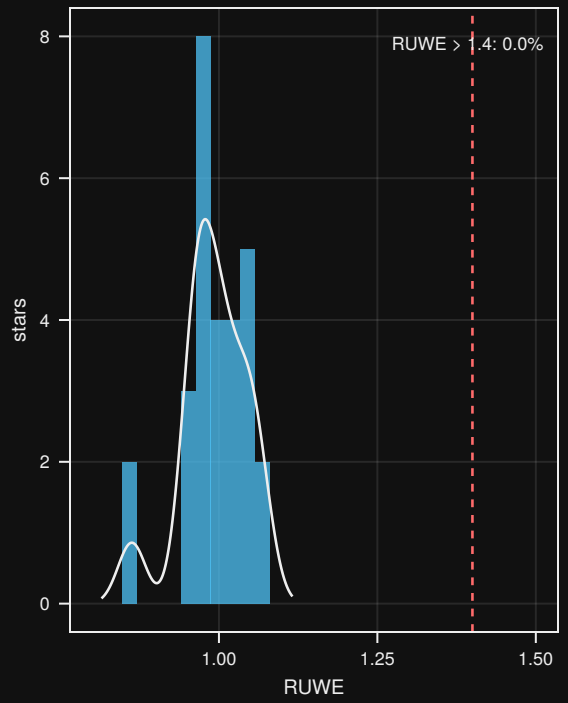
FLAME age (n = 8)



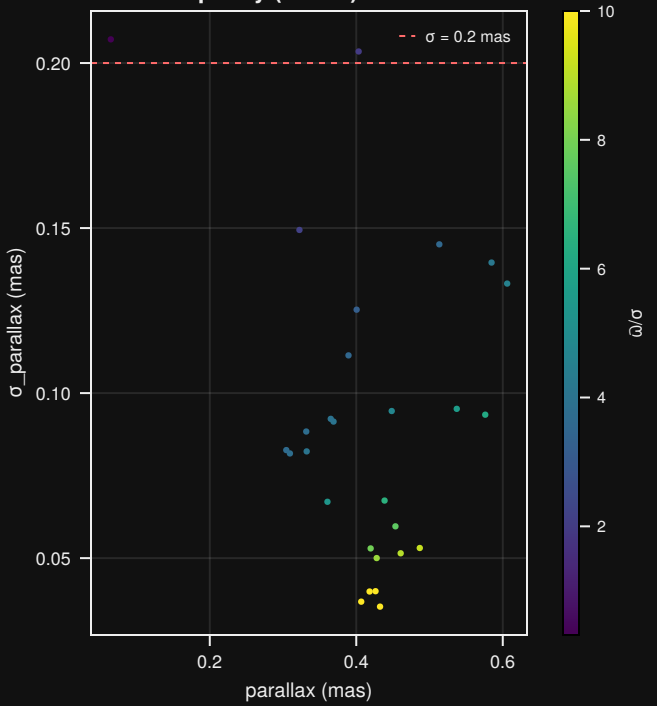
Age–metallicity (n = 8)



RUWE (n = 28)

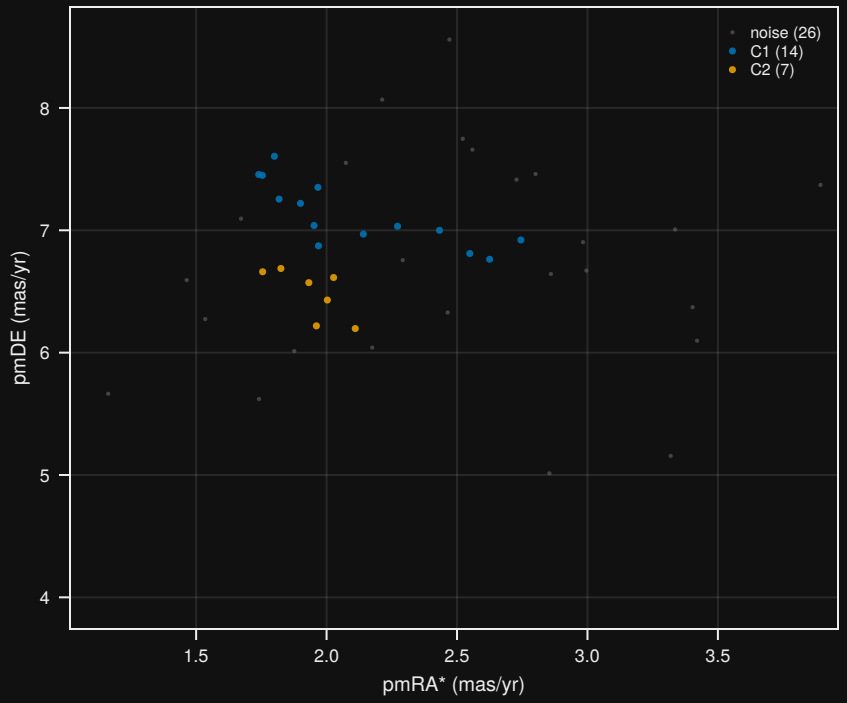


Parallax quality (n = 28)

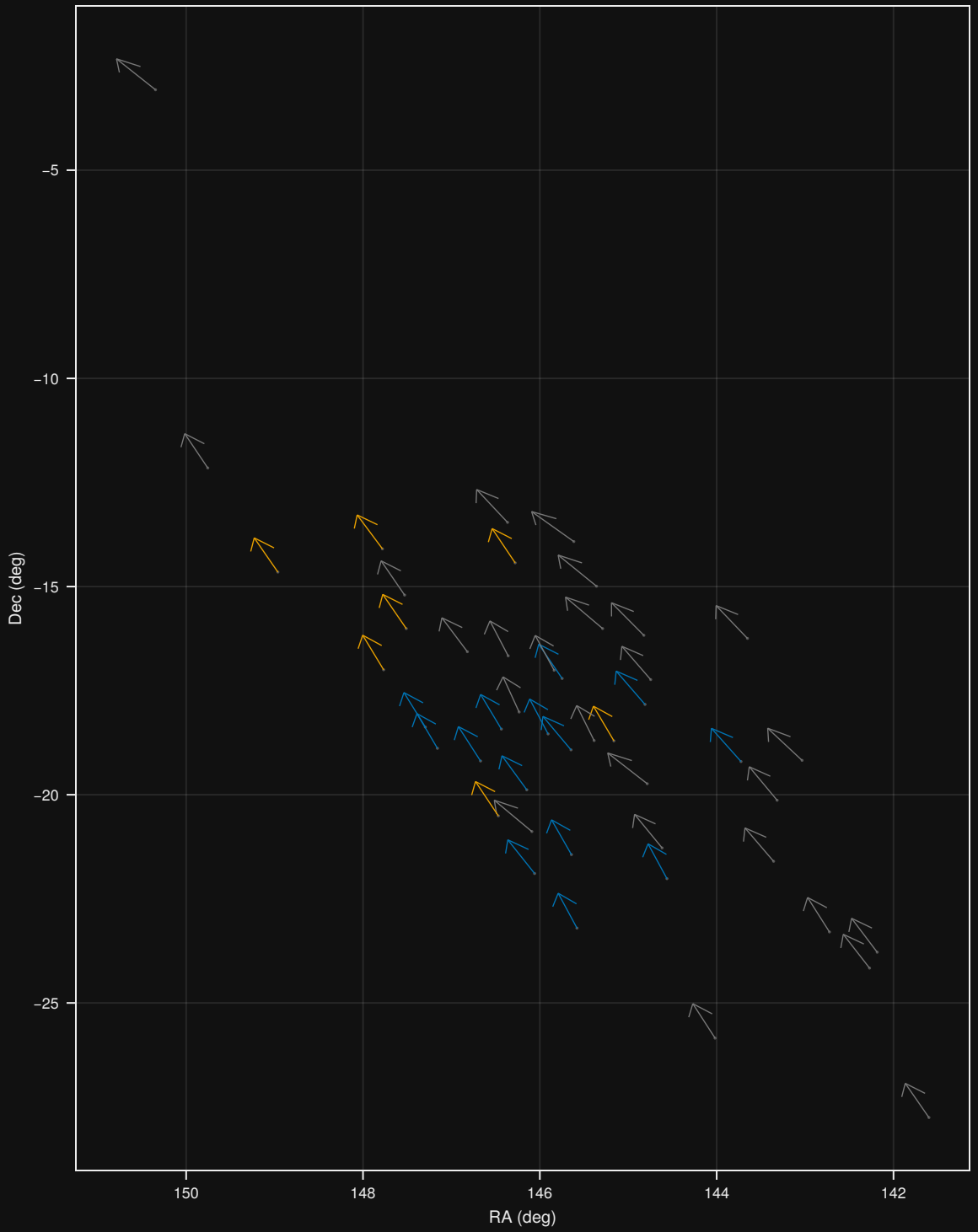


New-10

PM clustering · HDBSCAN · 2 cluster(s) · noise 55%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	26	2.50	6.66	0.67	-1.29	3
C1	14	1.97	7.04	0.29	-1.41	1
C2	7	1.96	6.57	0.10	-1.30	3

New-10

Per-cluster proper-motion vector maps

